

Name:

BUID:

More Practice Problems

1.

$$A = \begin{bmatrix} 0 & 2 & 0 & 4 & 1 \\ 0 & -3 & 1 & -3 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{bmatrix}$$

- (a) Determine a basis for $\text{Col } A$ (b) Determine a basis for $\text{Nul } A$
(c) Express $\mathbf{0}$ as a nontrivial linear combination of the columns of A
(d) If A is interpreted as the augmented matrix of a linear system, does this system have a unique least squares solution?

2. Let A be a 7×5 matrix such that

$$A^T A = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- (a) Determine the singular values of A (b) Determine $\dim(\text{Nul } A)$ using the fact that $\text{Nul } A = \text{Nul } A^T A$
(c) Determine $\text{rank } A$ (d) Determine $\text{rank } A^T$ and $\dim(\text{Nul } A^T)$
(e) Determine V in the SVD of A (assuming nonincreasing order of singular values in Σ)

3.

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \end{bmatrix} \right\}$$

- (a) Determine the matrix that implements the change of basis matrix for $\mathcal{B}$
(b) Determine an orthonormal basis of $\mathbb{R}^2$ which contains a scalar multiple of $\begin{bmatrix} -2 \\ -4 \end{bmatrix}$

4. Let A be a *symmetric* matrix such that $\text{Nul}(A) = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$ and $A \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$. Determine A

Write your solutions below and on the back on this page.

Continue your work here.